Rishabh Patidar

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Gwalior, Madhya Pradesh, India

Research Interests

Machine learning for structured and tabular data, with a focus on feature engineering, predictive modelling, and model interpretability. Applied generative AI and retrieval-augmented generation (RAG) systems for knowledge-intensive tasks. Multimodal AI systems combining vision, audio, and language signals for real-world behavioural analysis. Data-centric approaches to real-world business problems in emerging markets.

Education

Madhav Institute of Technology and Science Gwalior, MP
B.Tech in Information Technology | ***CGPA: 8.124*** *Expected: May 2027*
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Statistics, Linear Algebra, Computer Networks

Research Experience

Undergraduate Research Assistant Jan – May 2026
Madhav Institute of Technology and Science *Gwalior, MP*

- Conducted end-to-end exploratory data analysis on event marketing datasets covering 50+ features, identifying key footfall drivers through correlation analysis and domain-informed feature selection, reducing model noise by 30%
- Architected a Two-Stage XGBoost Regressor pipeline ($R^2 = 0.96$, RMSE = 215.6 vs 1,452.3 for baseline linear regression) by engineering venue and marketing features with cross-validated hyperparameter tuning via GridSearchCV
- Designed a Dynamic Physical Clamping algorithm introducing hard venue-capacity constraints at inference time, eliminating all physically invalid predictions and ensuring production-safe model outputs across 100% of test cases
- Integrated LLaMA-3.3-70B via Groq API with zero-trust key validation to perform semantic brand-event synergy scoring and generate structured JSON negotiation reports, deployed as a stateless FastAPI microservice
- Authored a full academic project report covering system architecture, ML pipeline design, model evaluation, cybersecurity protocols, and societal impact analysis; approved and signed by department faculty and Head of Department

Academic Reports and Submissions

Rishabh Patidar and Sujal Hammad. *"SponsorWise: An AI-Powered Event Sponsorship Intelligence Platform."* Minor Project Report, Department of Information Technology, Madhav Institute of Technology and Science, Gwalior. Supervised by Dr. Neha Bhardwaj. January – May 2026. *[Approved and submitted]*

Technical Projects

Code Archaeologist — Git History RAG System *2026*
Python, RAG, Pinecone, BGE Sentence Transformers, BM25, LLaMA-3.3-70B, Groq API, PyGithub, Redis, Streamlit

- Designed and deployed a production-ready retrieval-augmented generation system that ingests any public GitHub repository, extracts commits and diffs via GitPython, enriches chunks with PR metadata via PyGithub, and classifies each by developer intent (bug fix, feature, refactor, chore) before storing embeddings in Pinecone with per-user namespace isolation and TTL-based auto cleanup via Redis
- Implemented hybrid retrieval combining dense semantic search (BAAI/bge-small-en-v1.5 sentence transformers) with BM25 keyword reranking at 60/40 weight, improving precision for exact identifiers such as function names and file paths that pure semantic search misses
- Deployed on Streamlit Cloud with IP-based Redis rate limiting, admin access control, and a six-tab dashboard featuring co-change coupling detection, file evolution timelines, author ownership mapping, PR analysis, and LLM-generated commit summaries via Groq API

SponsorWise — Event Sponsorship Intelligence Platform *2026*
React 18, Node.js, Express.js, FastAPI, XGBoost, MongoDB, Socket.IO, Groq API

- Architected a production-ready full-stack SaaS platform on a distributed microservices model: a stateless Python FastAPI inference engine for ML computation, a stateful Node.js/Express backend for authentication and persistence, and a React 18 SPA frontend — each isolated via a Dual-Axios zero-trust routing layer
- Engineered a Two-Stage XGBoost pipeline ($R^2 = 0.96$) with a Dynamic Physical Clamping algorithm enforcing venue-capacity constraints at inference; integrated LLaMA-3.3-70B via Groq API to generate structured JSON negotiation battlecards and automated sponsor outreach emails from raw statistical outputs
- Implemented a real-time WebSocket negotiation layer using Socket.IO with isolated conversation rooms, atomic MongoDB synchronization, and a global notification engine; secured with dual-token JWT authentication, HTTP-only cookies, and Role-Based Access Control (RBAC)

Customer Segmentation & Churn Prediction *2025*

Scikit-learn, XGBoost, SHAP, SMOTE, Streamlit, SQL

- Engineered 22 RFM and behavioural features from 2.5M+ raw transactions via a modular SQL-to-Pandas ETL pipeline; applied KMeans (k=4) clustering to surface 4 actionable segments including a 15.7% churn-rate cohort
- Trained a Random Forest classifier (ROC-AUC 0.8793) with SMOTE oversampling; used SHAP TreeExplainer to confirm recency as the dominant churn driver and translated probability outputs into a $233,917 revenue-at-risk estimate across 392 flagged households

CinemaIQ — Box Office Intelligence Platform *2025*

Python, XGBoost, Scikit-learn, Streamlit, Plotly, Groq API

- Designed novel actor, director, and composer power-index features using weighted historical performance aggregation and multi-hot genre encoding; benchmarked 5 regression models selecting XGBoost (lowest RMSE, best holdout generalisation) for South Asian film revenue prediction
- Deployed a 3-tab Streamlit dashboard with Pessimistic/Base/Optimistic scenario simulation, ROI gauges, and LLaMA 3.3 AI commentary via Groq API, enabling end-to-end revenue forecasting from raw feature input to actionable insight

Video Target Identification System *2025*

Python, InsightFace, MediaPipe, OpenCV, Streamlit

- Built a biometric analytics pipeline fusing InsightFace 512-d ArcFace embeddings and MediaPipe 12-d pose/gait descriptors via weighted cosine similarity (face 0.7, pose 0.3) enabling person re-identification across multi-camera CCTV footage
- Deployed a Streamlit forensic dashboard supporting multi-video batch scanning, real-time similarity score visualisation, and automated evidence export (PDF report, CSV log, ZIP of face crops) for downstream investigative use

Technical Skills

Languages: Python, SQL, TypeScript, Java, C/C++
AI / LLM: Retrieval-Augmented Generation (RAG), Vector Embeddings, Sentence Transformers, Prompt Engineering, LLaMA-3.3-70B, Groq API, Hybrid Search (Dense + BM25), Pinecone, Redis, Audio Analysis (Whisper), Semantic Similarity Scoring, Voice Recognition
ML / Modelling: Scikit-learn, XGBoost, Random Forest, Gradient Boosting, KMeans, PCA, SHAP, SMOTE, RFM Analysis, Feature Engineering, Hyperparameter Tuning, A/B Testing, Hypothesis Testing
Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, Statsmodels, Excel
Web & Deployment: React 18, Node.js, Express.js, FastAPI, Pydantic V2, MongoDB, Socket.IO, Streamlit, REST APIs, WebSockets, JWT, RBAC
Tools: Git, GitPython, Docker (basic), Jupyter, Google Colab, Vercel, Cloudinary CDN
Computer Vision: InsightFace (ArcFace), MediaPipe (Face Mesh 468-point), OpenCV, face-api.js — applied in forensic analytics, behavioural analysis, and real-time posture detection

Certifications

Mathematical Foundations of Machine Learning | NPTEL, IIT Madras | Score: 73/100 | 2025

Achievements and Awards

1st Runner-Up — SSH '26 National Hackathon | SSH | Feb 2026

Top Performer, WebDev Track — Hacksagon | ABV IIITM Gwalior | Apr 2026

Finalist — ABV-IIITM Hackatron | ABV-IIITM (Powered by GitHub) | Oct 2025

Additional Information

Languages: English (Professional), Hindi (Native)
Interests: Applied ML research, data-centric AI, LLM systems, multimodal AI, open-source development
GitHub: github.com/RishabhPR77 — deployed projects with live demos